

Data Availability

A.1. Policy samples from Hangzhou and Shanghai

City	Code	Policy Title	Year	Issuing Authority
Hangzhou	P1	Hangzhou Municipal Several Policies for Establishing a National Pilot Zone for Innovative Development of Next-Generation Artificial Intelligence	2019	General Office of the Hangzhou Municipal People's Government
	P2	Implementation Opinions on Accelerating the Innovative Development of the Artificial Intelligence Industry	2023	General Office of the Hangzhou Municipal People's Government
	P3	Hangzhou Action Plan for High-Quality Development of the Full Artificial Intelligence Industry Chain (2024–2026)	2025	Hangzhou Municipal Bureau of Economy and Information Technology (Hangzhou Municipal Bureau of Digital Economy)
	P4	Several Measures to Support the High-Quality Development of the Full Artificial Intelligence Industry Chain	2024	General Office of the Hangzhou Municipal People's Government
	P5	Hangzhou 14th Five-Year Plan for the Development of the Artificial Intelligence Industry (2021–2025)	2021	Hangzhou Municipal Bureau of Economy and Information Technology, Hangzhou Municipal Development and Reform Commission
Shanghai	P6	Shanghai Implementation Opinions on Promoting the Development of Next-Generation Artificial Intelligence	2017	General Office of the Shanghai Municipal People's Government
	P7	Shanghai Implementation Plan for Promoting the “AI+” Initiative to Create “Smart and Easy” Government Services	2024	General Office of the Shanghai Municipal People's Government
	P8	Implementation Plan on Artificial Intelligence “Shaping Shanghai” (Mo Su Shen Cheng)	2024	General Office of the Shanghai Municipal People's Government
	P9	Shanghai Regulations on Promoting AI Industry Development	2022	15th Shanghai Municipal People's Congress Standing Committee

	P10	Shanghai Medical AI Development Work Plan (2025–2027)	2024	General Office of the Shanghai Municipal People’s Government
	P11	Shanghai Maqiao AI Innovation Pilot Zone Construction Work Plan	2019	General Office of the Shanghai Municipal People’s Government
	P12	Shanghai Action Plan on AI Empowering High-Quality Development of Basic Education (2024–2026)	2024	Shanghai Municipal Education Commission
	P13	Shanghai Several Measures to Promote Innovative Development of AI Large Models (2023–2025)	2023	Shanghai Municipal Commission of Economy and Informatization, Shanghai Municipal Development and Reform Commission, Shanghai Municipal Science and Technology Commission, Office of the Cyberspace Affairs Commission of the Shanghai Municipal Committee of the CPC, Shanghai Municipal Finance Bureau
	P14	Shanghai 14th Five-Year Plan for Artificial Intelligence Industry Development (2021–2025)	2021	Shanghai Municipal Commission of Economy and Informatization
	P15	Shanghai Guiding Opinions on Building the Next-Generation AI Standards System	2021	Shanghai Municipal Commission of Economy and Informatization, Shanghai Municipal Administration for Market Regulation
	P16	Shanghai New-Generation Artificial Intelligence Algorithm Innovation Action Plan (2021–2023)	2021	Shanghai Municipal Commission of Economy and Informatization, Shanghai Municipal Development and Reform Commission, Shanghai Municipal Education Commission, Shanghai Municipal Science and Technology Commission
	P17	Action Plan for Building Shanghai into an Artificial Intelligence Highland and Creating a World-Class Innovation Ecosystem (2019–2021)	2019	Shanghai Municipal Commission of Economy and Informatization
Table A.1. 17 AI Policies from Hangzhou and Shanghai				

A.2. Keyword TF-IDF Scores

The top 100 most important keywords are identified by calculating the TF-IDF scores.

Word	TF-IDF	Word	TF-IDF	Word	TF-IDF
AI	0.02789	Intelligence	0.02194	Innovation	0.01411
Utilization	0.01362	Technology	0.01341	Development	0.01255
Model	0.01107	Domain	0.01107	Scenario	0.01090
Research	0.00933	Priority	0.00928	Network	0.00900

Chip	0.00877	R&D	0.00848	Capability	0.00825
Algorithm	0.00816	Society	0.00815	Strategy	0.00786
Human-Machine	0.00757	Nation	0.00744	Holism	0.00723
Enterprise	0.00720	Indicator	0.00717	Product	0.00678
Base	0.00628	Objective	0.00626	Security	0.00613
Keystone	0.00608	Unit	0.00600	Situation	0.00560
Environment	0.00558	Management	0.00554	Level	0.00545
Information	0.00537	Industry	0.00536	Policy	0.00511
Organization	0.00506	Intelligentization	0.00483	Support	0.00482
Service	0.00473	Collaboration	0.00472	System	0.00471
Mechanism	0.00471	Wisdom	0.00470	Demonstration	0.00465
Project	0.00458	Talent	0.00458	Production	0.00451
University	0.00447	Foundation	0.00444	Task	0.00444
Resource	0.00442	Planning	0.00438	Integration	0.00438
Configuration	0.00437	Institution	0.00431	Infrastructure	0.00429
Open-Source	0.00426	Issue	0.00425	Theory	0.00425
Methodology	0.00424	Direction	0.00423	Evaluation	0.00418
Fundamental Theory	0.00411	Platform	0.00402	Education	0.00402
Achievement	0.00402	Data	0.00398	Breakthrough	0.00387
Science- and- Technology	0.00382	Cooperation	0.00381	Economy	0.00370
Plan	0.00369	Standard	0.00369	Talent Development	0.00365
Analysis	0.00361	Advantage	0.00361	Framework	0.00353
Industrialization	0.00352	Deployment	0.00352	Cultivation	0.00351
Internationality	0.00349	Sharing	0.00346	Pilot Zone	0.00345
Cross-Media	0.00341	Civil-Military	0.00340	Sector	0.00332
Knowledge	0.00329	Robot	0.00327	Automation	0.00324
Collective	0.00324	Ecosystem	0.00320	Generativity	0.00318
Orientation	0.00316	Decision-Making	0.00315	Monitoring	0.00314
Unmanned Systems	0.00314	Hybridization	0.00310	World	0.00308
Discipline	0.00307				
Table A.2. Top 100 Keywords Ranked by TF-IDF Scores in 103 Central-Level AI Policy Texts					

A.3. Model Parameters

Parameter	Value and Description
vector_size	150 (dimension of word vectors)
window	5 (context window size)
min_count	2 (minimum word frequency)
workers	6 (number of parallel processing threads)

Table A.3. Word2Vec Model Parameter Settings

Parameter	Value and Description
n_topics	(5, 50) (parameter search space)
random_state	42 (random seed)
init_points	5 (number of initial random exploration points)
n_iter	500 (number of optimization iterations)
Table A.4. Bayesian Optimization Parameters	

A.4. Bertopic Model Output

Theme 1, “Artificial Intelligence Standardization and Technical Norms”, centers on establishing unified technical standards and frameworks to facilitate technology transfer and cross-sector collaboration, emphasizing achievement transformation, safety verification, and infrastructure norms. Theme 2, “Intelligent Manufacturing and System Integration”, focuses on developing new intelligent equipment and systems, particularly industrial robots with self-detection and self-calibration capabilities, as well as deep learning-based intelligent inspection and logistics systems. Theme 3, “Talent Cultivation and Artificial Intelligence Education”, revolves around reforming education models, integrating AI into curricula, establishing interdisciplinary “AI + X” training programs, and organizing competitions to build a robust talent pipeline. Theme 4, “Artificial Intelligence Governance and Risk Regulation”, underscores legal oversight, privacy and data security protection, risk assessment in ethical and employment domains, and forward-looking policies for safe and controllable development. Theme 5, “Application of Artificial Intelligence in Industry Scenarios”, highlights deep integration of AI with IoT, big data, and cloud computing in smart cities, intelligent agriculture, healthcare, finance, and precision resource management. Theme 6, “Artificial Intelligence Infrastructure and Computing Power Platforms”, concentrates on building high-speed 5G networks, AI chips, cloud platforms, and computing infrastructure. Theme 7, “International Cooperation and Global Artificial Intelligence Governance”, stresses joint international efforts in law, ethics, education, and resource sharing. Theme 8, “Fundamental Research and Theoretical Breakthroughs in Artificial Intelligence”, targets frontier basic research and paradigm-shifting computational architectures. Theme 9, “Public Participation and Popularization of Artificial Intelligence”, promotes societal engagement through exhibitions, experience centers, and public science activities. Theme 10, “Artificial Intelligence-Driven Economic and Social Transformation”, examines AI’s role in upgrading traditional industries and driving broader societal intelligent transformation. Theme 11, “Data Integration and Resource Sharing”, focuses on constructing standardized, high-quality shared data platforms and ecosystems. Finally, Theme 12, “Artificial Intelligence Innovation and Ecosystem Construction”, emphasizes establishing innovation bases and pilot zones, fostering regional cooperation, and building a collaborative government–enterprise–research ecosystem to foster multidimensional AI innovation.

Topic	Representative Content
Topic 0	Introduce policy instruments to promote AI commercialization and safety verification; clarify infrastructure standards and operational models; formulate industry standards for cross-sector collaboration.
Topic 1	Develop industrial robots with self-detection/correction; apply deep learning for intelligent inspection and advanced logistics equipment.
Topic 2	Integrate AI into teaching (high-quality courses/textbooks); establish “AI + X” talent model; organize AI education activities and competitions.
Topic 3	Establish norms for user-data protection in generative AI; strengthen risk assessment (ethics, employment); formulate forward-looking safety policies.

Topic 4	Leverage IoT, big data, cloud computing in smart cities and ecology; promote AI in healthcare, finance, transportation; explore precision agriculture applications.
Topic 5	Build 5G high-speed networks; develop AI chips, compilers, drivers; strengthen cloud computing and intelligent network infrastructure.
Topic 6	Strengthen international cooperation on AI legal frameworks and rules; promote connectivity in education, research, and industry.
Topic 7	Focus on foundational theory bottlenecks and paradigm shifts; explore hybrid augmented intelligence and new computing frameworks.
Topic 8	Establish AI exhibition halls and experience centres; organize public science activities to increase engagement.
Topic 9	Transform high-energy-consumption and labor-intensive industries with AI; drive industrial economy innovation and societal intelligentization.
Topic 10	Build national big-data platforms for forestry/grasslands/ecology; promote high-quality training-data standardization and open sharing.
Topic 11	Establish AI innovation bases and experimental platforms; promote demonstration zones and scenario-driven ecosystem collaboration.
Table A.5. Topic Identification Results	

A.5. Secondary Indicator Score Criterion

Secondary Indicator	Score Criterion
X1:1 forecasting	Determine whether the policy is predictive; if it is, score = 1; if not, score = 0
X1:2 recommendation	Determine whether the policy is advisory/recommendatory; if yes, score = 1; otherwise 0
X1:3 supervision	Determine whether the policy involves oversight/monitoring; if yes, score = 1; otherwise 0
X1:4 regulation	Determine whether the policy has binding regulatory force; if yes, score = 1; otherwise 0
X1:5 guidance	Determine whether the policy provides directional guidance; if yes, score = 1; otherwise 0
X1:6 planning	Determine whether the policy is a strategic or development plan; if yes, score = 1; otherwise 0
X2:1 temporary (within the current year)	Determine whether the policy duration is within the current year; if yes, score = 1; otherwise 0
X2:2 short-term (1–3 years)	Determine whether the policy duration is 1–3 years; if yes, score = 1; otherwise 0
X2:3 medium-term (3–5 years)	Determine whether the policy duration is 3–5 years; if yes, score = 1; otherwise 0
X2:4 long-term (>5 years)	Determine whether the policy duration exceeds 5 years; if yes, score = 1; otherwise 0
X2:5 Rolling	Determine whether the policy adopts rolling or unspecified long-term implementation; if yes, score = 1; otherwise 0
X3:1 government	Determine whether government departments are explicitly targeted; if yes, score = 1; otherwise 0
X3:2 enterprise	Determine whether enterprises are explicitly targeted; if yes, score = 1; otherwise 0

X3:3 university	Determine whether universities are explicitly targeted; if yes, score = 1; otherwise 0
X3:4 research institute	Determine whether research institutes are explicitly targeted; if yes, score = 1; otherwise 0
X3:5 social organization	Determine whether social organizations are explicitly targeted; if yes, score = 1; otherwise 0
X3:6 public	Determine whether the general public is explicitly targeted; if yes, score = 1; otherwise 0
X3:7 international organization	Determine whether international organizations are explicitly targeted; if yes, score = 1; otherwise 0
X4:1 AI Standardization and Technical Norms	Determine whether the policy addresses AI standardization/technical norms; if yes, score = 1; otherwise 0
X4:2 Intelligent Manufacturing and System Integration	Determine whether the policy addresses intelligent manufacturing/system integration; if yes, score = 1; otherwise 0
X4:3 Talent Cultivation and AI Education	Determine whether the policy addresses talent cultivation/AI education; if yes, score = 1; otherwise 0
X4:4 AI Governance and Risk Regulation	Determine whether the policy addresses AI governance/risk regulation; if yes, score = 1; otherwise 0
X4:5 Application of AI in Industry Scenarios	Determine whether the policy addresses industry-specific AI applications; if yes, score = 1; otherwise 0
X4:6 AI Infrastructure and Computing Power Platforms	Determine whether the policy addresses AI infrastructure/computing power; if yes, score = 1; otherwise 0
X4:7 International Cooperation and Global AI Governance	Determine whether the policy addresses international cooperation/global governance; if yes, score = 1; otherwise 0
X4:8 Fundamental Research and Theoretical Breakthroughs in AI	Determine whether the policy addresses fundamental AI research; if yes, score = 1; otherwise 0
X4:9 Public Participation and Popularization of AI	Determine whether the policy addresses public participation/AI popularization; if yes, score = 1; otherwise 0
X4:10 AI-Driven Economic and Social Transformation	Determine whether the policy addresses AI-driven economic/social transformation; if yes, score = 1; otherwise 0
X4:11 Data Integration and Resource Sharing	Determine whether the policy addresses data integration/resource sharing; if yes, score = 1; otherwise 0
X4:12 AI Innovation and Ecosystem Construction	Determine whether the policy addresses AI innovation/ecosystem construction; if yes, score = 1; otherwise 0
X5:1 legal protection	Determine whether the policy offers legal/intellectual-property protection; if yes, score = 1; otherwise 0
X5:2 financial support	Determine whether the policy offers financial subsidies/funds; if yes, score = 1; otherwise 0
X5:3 talent incentive	Determine whether the policy offers talent attraction/subsidy measures; if yes, score = 1; otherwise 0
X5:4 resource sharing	Determine whether the policy promotes resource/data sharing mechanisms; if yes, score = 1; otherwise 0
X5:5 technical support	Determine whether the policy provides technical guidance/platforms; if yes, score = 1; otherwise 0

X5:6 tax benefits	Determine whether the policy offers tax preferences/reductions; if yes, score = 1; otherwise 0
X6:1 scientific value	Determine whether the policy reflects scientific value; if yes, score = 1; otherwise 0
X6:2 technological value	Determine whether the policy reflects technological value; if yes, score = 1; otherwise 0
X6:3 economic value	Determine whether the policy reflects economic value; if yes, score = 1; otherwise 0
X6:4 social value	Determine whether the policy reflects social value; if yes, score = 1; otherwise 0
X6:5 cultural value	Determine whether the policy reflects cultural value; if yes, score = 1; otherwise 0
X6:6 environmental value	Determine whether the policy reflects environmental value; if yes, score = 1; otherwise 0
X7:1 voluntary	Determine whether implementation is voluntary only; if yes, score = 1; otherwise 0
X7:2 mandatory	Determine whether implementation is mandatory only; if yes, score = 1; otherwise 0
X7:3 mixed	Determine whether implementation is mixed (voluntary + mandatory); if yes, score = 1; otherwise 0
X8:1 detailed content	Determine whether the policy content is specific and detailed; if yes, score = 1; otherwise 0
X8:2 scientific planning	Determine whether the policy exhibits scientific and reasonable planning; if yes, score = 1; otherwise 0
X8:3 clear objective	Determine whether the policy objectives are clear and explicit; if yes, score = 1; otherwise 0
X8:4 clear responsibility	Determine whether responsibilities of all parties are clearly assigned clearly; if yes, score = 1; otherwise 0
X8:5 stakeholder participation	Determine whether multiple stakeholders are encouraged to participate; if yes, score = 1; otherwise 0
X9:1 emergency management	Determine whether emergency management is an explicit objective; if yes, score = 1; otherwise 0
X9:2 risk prevention	Determine whether risk prevention/control is an explicit objective; if yes, score = 1; otherwise 0
X9:3 public service	Determine whether improving public service is an explicit objective; if yes, score = 1; otherwise 0
X9:4 economic development	Determine whether promoting economic development is an explicit objective; if yes, score = 1; otherwise 0
X9:5 scientific and technological innovation	Determine whether fostering S&T innovation is an explicit objective; if yes, score = 1; otherwise 0
X9:6 environmental protection	Determine whether environmental protection is an explicit objective; if yes, score = 1; otherwise 0
X9:7 national security	Determine whether safeguarding national security is an explicit objective; if yes, score = 1; otherwise 0
X9:8 resource allocation	Determine whether optimizing resource allocation is an explicit objective; if yes, score = 1; otherwise 0

X9:9 international competitiveness	Determine whether enhancing international competitiveness is an explicit objective; if yes, score = 1; otherwise 0
Table A.6. Secondary Indicator Score Criterion	

A.6. Results of the Multi-Input-Output Table for Policy Samples

P	X1						X2					X3			
	X1:1	X1:2	X1:3	X1:4	X1:5	X1:6	X2:1	X2:2	X2:3	X2:4	X2:5	X3:1	X3:2	X3:3	X3:4
P1	0	0	1	1	1	1	0	0	1	0	0	1	1	1	1
P2	0	0	1	1	1	1	0	1	0	0	0	1	1	1	1
P3	1	0	1	1	1	1	0	1	0	0	0	1	1	1	1
P4	0	0	1	1	1	1	0	1	0	0	0	1	1	1	1
P5	1	1	1	1	1	1	0	1	1	0	1	1	1	1	1
P6	1	0	1	1	1	1	0	1	0	1	0	1	1	1	1
P7	0	0	1	1	1	1	0	1	0	0	0	1	1	0	0
P8	0	0	1	1	1	1	0	1	0	0	0	1	1	1	1
P9	0	0	1	1	1	1	0	0	0	1	0	1	1	1	1
P10	1	0	1	1	1	1	0	1	0	0	0	1	1	1	1
P11	1	0	1	0	1	1	0	1	1	1	0	1	1	1	1
P12	1	1	1	1	1	1	0	1	0	0	1	1	1	1	1
P13	1	1	1	1	1	1	0	1	0	0	0	1	1	1	1
P14	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1
P15	0	1	1	1	1	1	0	1	0	1	1	1	1	1	1
P16	0	1	0	1	1	1	0	1	0	0	1	1	1	1	1
P17	0	1	1	1	1	1	0	1	0	0	0	1	1	1	1
P	X3			X4											
	X3:5	X3:6	X3:7	X4:1	X4:2	X4:3	X4:4	X4:5	X4:6	X4:7	X4:8	X4:9	X4:10	X4:11	X4:12
P1	1	0	0	1	1	1	1	1	1	0	1	0	1	1	1
P2	1	0	0	1	1	1	1	1	1	0	1	0	1	1	1
P3	1	0	0	1	1	1	1	1	1	1	1	0	1	1	1
P4	0	0	0	0	1	1	0	1	1	0	1	0	1	0	1
P5	0	1	0	1	1	1	1	1	1	1	1	0	0	1	1
P6	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1
P7	0	1	0	1	0	0	1	1	1	0	0	0	0	1	1
P8	0	0	1	1	1	1	1	1	1	1	1	0	1	1	1
P9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
P10	0	0	0	1	0	1	1	1	1	0	1	0	0	1	1
P11	0	0	0	0	1	1	0	1	1	0	1	0	1	1	1
P12	1	0	0	1	0	1	1	1	0	0	1	0	0	1	0
P13	0	0	0	1	1	1	1	1	1	1	1	0	0	1	1
P14	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1
P15	1	0	0	1	1	1	1	1	1	1	0	0	1	1	1
P16	0	0	0	1	1	1	1	1	1	1	1	0	0	1	1
P17	1	0	0	1	1	1	1	1	1	1	1	0	1	1	1
P	X5						X6						X7		
	X5:1	X5:2	X5:3	X5:4	X5:5	X5:6	X6:1	X6:2	X6:3	X6:4	X6:5	X6:6	X7:1	X7:2	X7:3
P1	1	1	1	1	1	0	1	1	1	1	0	0	1	1	1

P2	0	1	1	1	1	0	1	1	1	1	0	1	1	1	1
P3	1	1	1	1	1	0	1	1	1	1	0	1	1	1	1
P4	0	1	1	1	1	0	1	1	1	0	0	0	1	1	1
P5	1	1	1	1	1	1	1	1	1	1	0	1	1	0	1
P6	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
P7	1	0	0	1	1	0	0	1	1	1	0	0	0	1	1
P8	0	1	1	1	1	0	1	1	1	1	1	1	1	0	1
P9	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1
P10	1	1	1	1	1	0	1	1	1	1	1	0	1	1	1
P11	0	1	1	1	1	0	1	1	1	1	1	1	1	0	1
P12	1	1	1	1	1	0	1	1	0	1	0	0	1	0	1
P13	1	1	1	1	1	0	1	1	1	1	0	0	1	1	1
P14	1	1	1	1	1	0	1	1	1	1	0	1	1	0	1
P15	1	1	1	1	1	0	1	1	1	1	0	0	1	0	1
P16	1	1	1	1	0	0	1	1	1	1	0	0	1	0	1
P17	1	1	1	1	1	1	1	1	1	1	0	0	1	0	1
P	X8					X9									
	X8:1	X8:2	X8:3	X8:4	X8:5	X9:1	X9:2	X9:3	X9:4	X9:5	X9:6	X9:7	X9:8	X9:9	
P1	1	1	1	1	1	0	1	1	1	1	0	1	1	0	
P2	1	1	1	1	1	1	1	1	1	1	1	1	1	0	
P3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
P4	1	1	1	1	1	0	1	0	1	1	0	0	0	0	
P5	1	1	1	1	1	1	1	1	1	1	1	0	1	1	
P6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
P7	1	1	1	1	1	0	1	1	1	1	0	0	1	0	
P8	1	1	1	1	1	1	1	1	1	1	1	0	1	1	
P9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
P10	1	1	1	1	1	1	1	1	1	1	0	0	1	0	
P11	1	1	1	1	1	0	0	1	1	1	1	0	1	0	
P12	1	1	1	1	1	0	1	0	0	1	0	0	1	0	
P13	1	1	1	1	1	0	1	1	1	1	0	0	1	1	
P14	1	1	1	1	1	0	1	1	1	1	1	0	1	1	
P15	1	1	1	1	1	0	1	1	1	1	0	0	1	1	
P16	1	1	1	1	1	0	1	1	1	1	0	0	1	1	
P17	1	1	1	0	1	0	1	1	0	1	0	0	1	1	
Table A.7. Multi-Input-Output Results of 17 Policies															

A.7. Results of the Multi-Input-Output Table for Policy Samples

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17
X1 Policy Type	0.67	0.67	0.83	0.83	1.00	0.83	0.67	0.67	0.67	0.83	0.83	1.00	1.00	1.00	0.83	0.67	0.83
X2 Policy Duration	0.20	0.20	0.20	0.20	0.60	0.40	0.20	0.20	0.20	0.20	0.60	0.40	0.20	0.60	0.60	0.40	0.20
X3 Policy Target	0.71	0.71	0.71	0.57	0.71	0.57	0.43	0.71	1.00	0.57	0.57	0.71	0.57	0.86	0.71	0.57	0.71

X4 Policy Content	0.83	0.83	0.92	0.58	0.83	0.92	0.50	0.92	1.00	0.67	0.67	0.50	0.83	0.92	0.83	0.83	0.92
X5 Policy Incentive	0.83	0.67	0.83	0.67	1.00	0.83	0.50	0.67	1.00	0.83	0.67	0.83	0.83	0.83	0.83	0.67	1.00
X6 Policy Function	0.67	0.83	0.83	0.50	0.83	1.00	0.50	1.00	0.83	0.83	1.00	0.50	0.67	0.83	0.67	0.67	0.67
X7 Policy Implementation	1.00	1.00	1.00	1.00	0.67	1.00	0.67	0.67	1.00	1.00	0.67	0.67	1.00	0.67	0.67	0.67	0.67
X8 Policy Design	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80
X9 Policy Objectives	0.67	0.89	1.00	0.33	0.89	1.00	0.56	0.89	1.00	0.67	0.56	0.33	0.67	0.78	0.67	0.67	0.56
Total	6.58	6.80	7.32	5.68	7.53	7.55	5.03	6.73	7.70	6.60	6.57	5.94	6.77	7.49	6.81	6.15	6.36
Table A.8. PMC Index Results of 17 Policies																	

A.8. Average Secondary Indicator Scores of Hangzhou and Shanghai

P	X1						X2					X3			
	X1:1	X1:2	X1:3	X1:4	X1:5	X1:6	X2:1	X2:2	X2:3	X2:4	X2:5	X3:1	X3:2	X3:3	X3:4
Hangzhou	0.40	0.20	1.00	1.00	1.00	1.00	0.00	0.80	0.40	0.00	0.20	1.00	1.00	1.00	1.00
Shanghai	0.42	0.50	0.92	0.92	1.00	1.00	0.00	0.83	0.17	0.42	0.42	1.00	1.00	0.92	0.92
P	X3			X4											
	X3:5	X3:6	X3:7	X4:1	X4:2	X4:3	X4:4	X4:5	X4:6	X4:7	X4:8	X4:9	X4:10	X4:11	X4:12
Hangzhou	0.60	0.20	0.00	0.80	1.00	1.00	0.80	1.00	1.00	0.40	1.00	0.00	0.80	0.80	1.00
Shanghai	0.42	0.17	0.25	0.92	0.75	0.92	0.92	1.00	0.92	0.67	0.83	0.08	0.50	1.00	0.92
P	X5						X6						X7		
	X5:1	X5:2	X5:3	X5:4	X5:5	X5:6	X6:1	X6:2	X6:3	X6:4	X6:5	X6:6	X7:1	X7:2	X7:3
Hangzhou	0.60	1.00	1.00	1.00	1.00	0.20	1.00	1.00	1.00	0.80	0.00	0.60	1.00	0.80	1.00
Shanghai	0.83	0.92	0.92	1.00	0.92	0.17	0.92	1.00	0.92	1.00	0.17	0.25	0.92	0.33	1.00
P	X8					X9									
	X8:1	X8:2	X8:3	X8:4	X8:5	X9:1	X9:2	X9:3	X9:4	X9:5	X9:6	X9:7	X9:8	X9:9	
Hangzhou	1.00	1.00	1.00	1.00	1.00	0.60	1.00	0.80	1.00	1.00	0.60	0.60	0.80	0.40	
Shanghai	1.00	1.00	1.00	0.92	1.00	0.33	0.92	0.92	0.75	1.00	0.25	0.08	1.00	0.75	

Table A.9. Average Secondary Indicator Scores of Hangzhou and Shanghai**A.9. Statistical Validation of Inter-City Differences**

These tests indicate that the small observed difference in average PMC scores (0.14 points) is not statistically meaningful and is likely attributable to sampling variation. Given the small and uneven sample sizes ($n=5$ and $n=12$), statistical power is limited, reducing the ability to detect subtle differences if they exist. These results support interpreting inter-city variations primarily through qualitative differences in individual PMC dimensions rather than overall scores.

City	N	Mean	Standard Deviation (SD)	95% Confidence Interval
Hangzhou	5	6.782	0.725	[5.881, 7.683]
Shanghai	12	6.642	0.747	[6.167, 7.116]
Table A.10. Descriptive Statistics				

Normality			
City	W Statistic	p-value	Interpretation
Hangzhou	0.942	0.682	Normal ($p>0.05$)
Shanghai	0.943	0.538	Normal ($p>0.05$)
Equality of Variances			
Test Statistic (F)	p-value	Interpretation	
0.004	0.953	Variances equal ($p>0.05$)	
Table A.11. Assumption Checks for Parametric Testing			

t-Statistic	Degrees of Freedom	p-value (two-tailed)	Conclusion
0.360	7.76	0.728	No significant difference ($p>0.05$)
Table A.12. Two-Sample t-Test			

U-Statistic	p-value (two-tailed)	Conclusion
33.0	0.799	No significant difference (confirm t-test; $p>0.05$)
Table A.13. Robustness Check: Mann–Whitney U Test (Non-Parametric)		

A.10. PMC-Surfaces of 17 Policies



